

What decision systems do

Power BI Service: Connect insights to delivery, alerts, and action - not just display

Choose the right system type

Deliver Insights

Push data to users automatically

- ✓ Standard subscriptions on a schedule
- ✓ Dynamic subscriptions per recipient
- ✓ PDF and PowerPoint attachments
- ✓ Data alerts when thresholds are crossed

Enable Self-Service

Let users explore independently

- ✓ Shared certified semantic models
- ✓ Personalize visuals on published reports
- ✓ Field parameters for dimension switching
- ✓ Analyze in Excel with live data

Connect to Action

Bridge insight and operation

- ✓ Power Automate for workflow triggers
- ✓ Power Apps visual for data write-back
- ✓ Embedded buttons to create tickets
- ✓ Budget inputs saved back to source

Key questions before building a decision system

Do users need to be pushed insights?

Do different users need different views?

Can users explore data independently?

Does the insight require an action?

Is the semantic model reusable and trusted?

What happens after the user sees the report?

"The goal is not to automate every decision. It is to reduce the distance between seeing an insight and acting on it."

Introduction: From Reports to Decisions

A report can be accurate, visually clear, secure, and fast - but still provide limited value when users must remember to open it, interpret the result, and manually decide what happens next. That is the gap between a **report** and a **decision system**.

In this series, a decision system means a Power BI solution that does more than display historical information. It uses reusable data assets, trusted business logic, automated delivery, alerts, and integrated actions to help people make and carry out decisions consistently. The journey can be understood as:



The Three Stages of Power BI Maturity

Stage 1: Traditional Report

A single report built for a single request (e.g., "Show me Q3 sales in Excel/Power BI"): It gets created, viewed once, abandoned, or duplicated by another team. This creates data silos, inconsistent metrics, and continuous maintenance overhead. The main limitation is not the report itself. It is that the report was treated as a one-time output rather than part of a reusable analytical system.

Stage 2: Report as a Reusable Data Asset

The next stage is to treat semantic models, transformation logic, measures, and reports as long-term business assets. A shared semantic model can centralise:

- Relationships and schema design
- Standard business definitions

- Reusable DAX measures
- Row-level security
- Date and calculation logic
- Trusted KPIs

Power BI semantic models provide reporting-ready data that can support multiple reports. With Build permission, authorised users can create additional reports and analytical content from the same model without reproducing the underlying logic.

Stage 3: Decision System

A decision system connects insights to delivery, review, or action. Instead of requiring users to continually check a report, the system may:

- Send a scheduled performance update
- Notify someone when a threshold is crossed
- Trigger a workflow from a report
- Allow users to record an action through an embedded app
- Provide governed data for independent analysis
- Distribute personalised report views to different recipients

The report becomes part of the organisation's operating process rather than a separate destination.

Reusable Power BI Data Assets

Types of Power BI Data Assets:

Power BI asset	What it provides	Reusability
Shared semantic model	Central relationships, measures, definitions, and security	High - can support many reports
Power BI report	Interactive, multi-page analysis built on a semantic model	Medium - can be distributed or embedded
Dashboard	A single monitoring surface containing pinned tiles	Medium - useful for high-level monitoring
Dataflow	Reusable cloud-based data preparation and transformation logic	High - prepared data can support multiple models
Power BI app	Curated and secure delivery of reports and dashboards	High - provides a consistent user experience

The key asset is often the semantic model. Managed self-service BI works best when organisations reuse a smaller number of governed semantic models rather than creating a separate model for every report.

Types of Power BI Decision Systems

A decision system takes static data assets and turns them into **active workflows** that prompt actions or streamline operational choices.

1. Alerting & Exception Systems

- **How it works:** Instead of checking a report daily, users only get notified when something goes wrong or hits a threshold.
- **Power BI Feature:** Power BI Data Alerts + Power Automate.
- **Example:** Sending an urgent Microsoft Teams message to a warehouse manager the second stock levels for an item drop below 10%.

2. Cadence & Review Systems (e.g., Weekly/Monthly Reports)

- **How it works:** Automated, scheduled delivery of key insights mapped to regular business rhythms (like executive standups or board meetings).
- **Power BI Feature:** Scheduled Email Subscriptions + Export to PowerPoint/PDF + Paginated Reports.
- **Example:** Every Monday at 8:00 AM, the regional leadership team automatically receives a PDF summary of the prior week's sales performance directly in their inbox.

3. Automated & Integrated Systems

- **How it works:** Data inside the report triggers external IT and operational systems automatically.
- **Power BI Feature:** Power Automate Visuals directly inside Power BI.
- **Example:** A manager notices an anomaly in a report, clicks a button embedded directly on the Power BI report page, and it automatically creates a ticket in ServiceNow or Jira for investigation.

4. Interactive "Write-Back" Decision Systems

- **How it works:** Allows users to input data, record decisions, or adjust assumptions directly inside the report UI, updating the underlying source system.
- **Power BI Feature:** Power Apps Visual embedded in Power BI.
- **Example:** A finance planner looks at Q4 projections in Power BI, adjusts the budget numbers using an embedded form, and saves it straight back to the SQL database without leaving the report.

Subscriptions in Power BI

What Are Subscriptions? : Subscriptions allow users to **receive automated email updates** of key Power BI reports and dashboards on a recurring schedule. Instead of opening the Power BI Service manually to check numbers every morning, Power BI sends a snapshot straight to your inbox.

Key Features

- **Custom Schedules:** Receive emails daily, weekly, monthly, or whenever data refreshes.
- **Email Attachments:** Include a preview image, PDF, or PowerPoint export of the report page.
- **Direct Links:** Includes a button in the email to jump straight to the live, interactive report in Power BI.
- **Dynamic Subscriptions:** Send personalized report views to different recipients based on recipient-specific data filters.

1. Standard Subscriptions

- **What it is:** Sends the exact same snapshot, page view, or PDF/PowerPoint attachment to everyone on the recipient list.
- **How it works:** You pick a scheduled time (e.g., *Daily at 8 AM*), enter the email addresses, and Power BI sends the static report or dashboard view.
- **Best used for:** Enterprise-wide broadcasts or team updates where everyone needs to see the exact same numbers (e.g., *Company-wide Monthly Financial Summary*).

2. Dynamic (Per-Recipient) Subscriptions

- **What it is:** Sends a personalized, filtered version of the report to each recipient based on who they are.
- **How it works:** Power BI uses an underlying mapping dataset (containing email addresses and their specific filter criteria). When the subscription runs, it dynamically applies those filters before emailing the report.
- **Best used for:** Scaled delivery across regions or managers. *Example:* Rather than creating 50 separate subscriptions, one dynamic subscription can send the **US report** to the US manager, the **UK report** to the UK manager, and the **Japan report** to the Japan manager automatically.

Self-Service BI Enablement

What Is Self-Service BI? Self-Service BI empowers non-technical business users (finance, marketing, sales) to **explore data, build custom visuals, and answer their own business questions** without relying on IT or report developers to build every visual for them.

How to Enable It in Power BI (Features & Methods)

1. **Shared Datasets / Semantic Models:** IT/Analytics creates and certifies *one reusable master data model*. Business users connect to this dataset to build their own custom reports without touching raw code or complex data cleaning.
2. **Personalize Visuals:** A report setting that lets end-users change chart types, swap axes, or add filters on a published report **on the fly**, saving their own version as a bookmark without breaking the original report.
3. **Field Parameters:** Developers add options so end-users can dynamically swap dimensions (e.g., toggle a chart view between *Region*, *Product*, or *Sales Rep*) using simple slicers.
4. **Analyze in Excel:** Business users who prefer Excel can connect directly to a published Power BI dataset and run PivotTables using live, centralized company data.

Concept	What it solves	Main benefit
Subscriptions	Users may forget or not have time to check reports	Delivers insights automatically
Dynamic subscriptions	Different users need different filtered views	Scales personalised distribution
Self-service BI	Users need to answer additional questions independently	Reduces the reporting bottleneck
Power Automate integration	Users need to act on an insight	Connects reports to workflows
Power Apps integration	Users need to input or approve information	Connects analysis with operational input



Practical Way to Decide

Choose the right Power BI reporting and action option for the outcome you need.

1 TRADITIONAL REPORT		Use a traditional Power BI report when: Users need interactive analysis, but no automated delivery or action is required.
2 SHARED SEMANTIC MODEL		Use a shared semantic model when: Several reports need the same business definitions, relationships, measures, and security rules.
3 POWER BI APP		Use a Power BI app when: A wider audience needs a curated and stable reporting experience.
4 STANDARD SUBSCRIPTION		Use a standard subscription when: All recipients need the same report view on a recurring schedule.
5 DYNAMIC SUBSCRIPTION		Use a dynamic subscription when: Recipients need personalised views based on region, department, account, or responsibility.
6 DATA ALERTS		Use data alerts when: Users only need to be notified when a supported KPI crosses a threshold.
7 POWER AUTOMATE VISUAL		Use the Power Automate visual when: A report selection should trigger an external action or workflow.
8 POWER APPS VISUAL		Use the Power Apps visual when: Users need to submit, approve, or update information from within the reporting experience.
9 PERSONALISED VISUALS		Use personalised visuals when: Readers need limited freedom to adjust a report without changing the published version.
10 FIELD PARAMETERS		Use field parameters when: Users need to switch measures or dimensions through controlled report options.
11 ANALYZE IN EXCEL		Use Analyze in Excel when: Business users prefer PivotTables but should continue working from trusted Power BI data.
12 DECISION SYSTEM		Use a decision system when: The business needs a repeatable flow from insight to review, alert, action, or recorded decision.



Turn reporting into action. Deliver the right experience for the right audience.



Conclusion

Decision systems represent the final layer of the Power BI journey.

A report becomes more valuable when it is built on reusable semantic models, distributed through a trusted channel, delivered at the right time, and connected to a clear action. Subscriptions push insights to users, self-service BI enables independent exploration, Power Automate connects reports to workflows, and Power Apps can bring operational input into the analytical experience. The objective is not to automate every business decision. It is to reduce the distance between **seeing an insight and acting on it**.

Across this 12-week series, we moved from business requirements and data connection through Power Query, data quality, modelling, DAX, report design, interactive storytelling, advanced analytics, performance, and governance.

The complete journey is:

Business question → Trusted data → Reliable model → Clear insight → Governed access → Informed action

That is how Power BI moves beyond dashboards and becomes part of a repeatable, trusted decision system.